THE CATHOLIC DIOCESE OF SAME
KINDOROKO GIRLS' SECONDARY SCHOOL
FORM FOUR LAST PRE–MOCK EXAMINATIONS

041 **BASIC MATHEMATICS**

(For Both Private and School Candidates)

Time: 2:30 Hours **Year: 2024**

Instructions

1. This paper consists of sections A and B with a total of **fourteen (14)** questions.
2. Answer **all** questions in all sections.
3. Each question in section A carries **six (06)** marks while each question in section B carries **ten (10)** marks
4. All necessary working and answers for each question must be shown clearly.
5. NECTA mathematical tables and non – programmable calculators may be used where necessary.
6. All communication devices and any unauthorized materials are **not** allowed in the examination room.
7. Write your **Examination Number** on every page of your answer booklet(s).

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KINDOROKO PRE MOCK 3 2024

SECTION A (60 Marks)

Answer **all** questions in this section

1. (a) (i) Traffic light at Kwamathuku junction in Ngujini street allows pedestrians after every 30 minutes. Another traffic light at Kwamaster junction in Kisangara street allows pedestrians after 45 minutes. If they start together to allow pedestrians at 6:00am, at what time will they allow pedestrians together again?

   (ii) Write 679.3494 correct to tenths.

   (b) Simplify the expression without using mathematical table, expressing the answer correct to two significant figures.

$$\frac{0.25 \times 8.85 \times 25}{0.00625 \times 0.5}$$

2. (a) Find the values of x and y that satisfy the equations:

$$\begin{cases} 5^{(x-2y)} = 25 \\ 3^{2x} \div 3^{y} = 3^{4} \end{cases}$$

   (b) Find the value of x if $\frac{logx}{log3} = \frac{log4}{log16}$

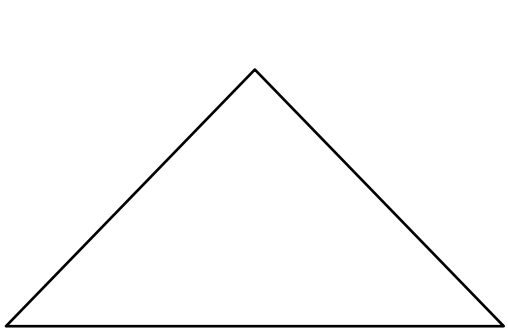
3. (a) In a school of 50 teachers, some drink Fanta and some drink Coca-cola. If 40 teachers drink Fanta, 18 drink Coca-cola and 14 drink both. How many teachers drink neither Fanta nor coca-cola?

   (b) Mama-p has two shirts, blue and red. She also has three skirts, black, green and yellow. By using a tree diagram, find the probability that she will put on a blue shirt and black skirt.

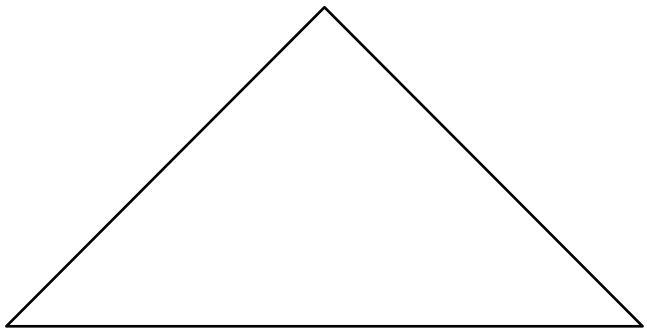
4. (a) Show that the triangle whose vertices are A(4, -4), B(-6, -2) and C(2,6) is an isosceles.

   (b) Find the equation of a perpendicular bisector of a line segment with ends at x-intercept 4 and y-intercept 6. Write the equation in the form of ax + by + c = 0

5. (a) The following triangles are similar. A side of small triangle is 6cm long and its corresponding side of the larger triangle is 20cm. If the area of the small triangle is 90cm^2, what is the area of the larger triangle?

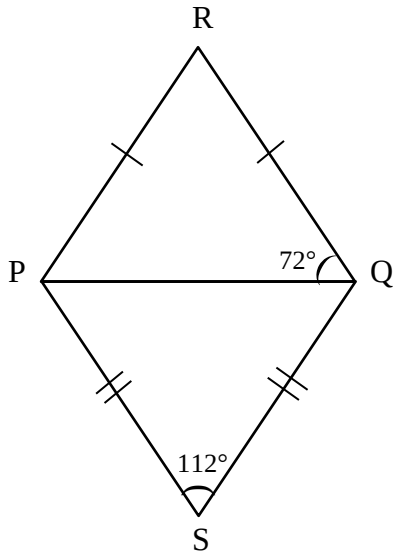


6cm



20cm

(b) In the following figure, angle RPQ = angle PQR and Angle SPQ = angle SQP. Find the size of the angle RPS.



6. (a) A motorcycle owned by Mathuku wa mathuku the Rice farmer at Kisangara village can travel from Mwanga to Moshi town 240 kilometers using 4 litres of petrol. Determine cost of petrol required to travel from Mwanga to Same 90 kilometers if one litre of petrol costs 2925 Tanzanian shillings.

   (b) The surface area of a circular object varies directly as the square of its radius. If its surface area is 78.5cm^2, when the radius is 5cm, find the surface area of the circular object when the radius is 7cm.

7. (a) Mr. Seth and Heavenlight shared 64,000 shillings in the ratio 3:5 respectively. Find the difference between their shares.

   (b) Determine the gross profit and net profit from the following information of Mr. Manimba who is a businessman at Kilomeni village in Mwanga District**.**

Sales ……	38,000/=
Opening stock ……..	8,000/=
Purchases …….	25,000/=
Electricity ……	4,000/=
Discount allowed ……..	2,000/=
Closing stock ……	5,000/=

8. (a) The sum of the first two terms of a geometrical progression is 10, and the sum of the first four terms is 40. Given that all terms of the progression are positive, show that:

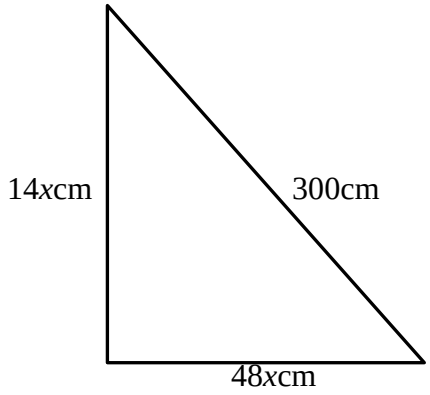
   (i) The common ratio is $\sqrt{3}$

   (ii) The sum of the first n terms is $5(3^{n/2} - 1)$

   (b) The second, fifth and seventh term of an arithmetic progression forms the first three consecutive terms of a geometric progression. Find the common ratio of the geometrical progression.

9. (a) A and B are two points on the ground level and both lie west of flagstaff. The angle of elevation of the top of the flagstaff from A is 56^0 and from B is 43^0. If B is 28m from the foot of the flagstaff, how far apart are the points A and B?

(b) The right-angled triangle in the diagram below has sides of length 14*x*cm, 48*x*cm, and 300cm. Find the value of *x*.



10. (a) The difference between two positive numbers is 7. If their product is 30, find the numbers.

(b) If $m = x^2$, find the values of x from the equation $x^4 - 13x^2 + 36 = 0$

SECTION B (40 Marks)

Answer **all** questions in this section

11. (a) In a certain Basic mathematics test for form four conducted at Kindoroko girls' secondary school. The following marks were awarded:

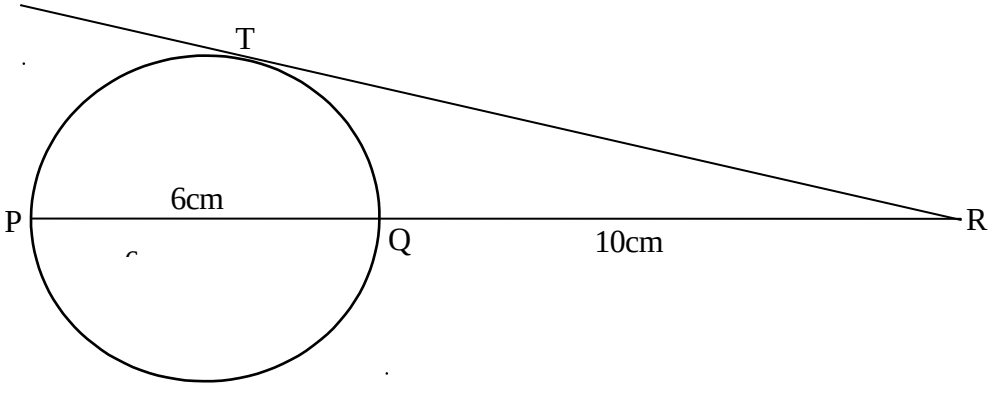
48	47	67	73	50	76	47	44
44	42	49	34	35	46	89	37
47	54	45	60	58	57	56	58
66	54	67	45	45	42	43	54
71	62	48	32	64	64	52	44

Prepare a frequency distribution table starting with 32 having class size of 10, and then find:

(i) Mean by assumed mean of 56.5

(ii) Mode of the distribution.

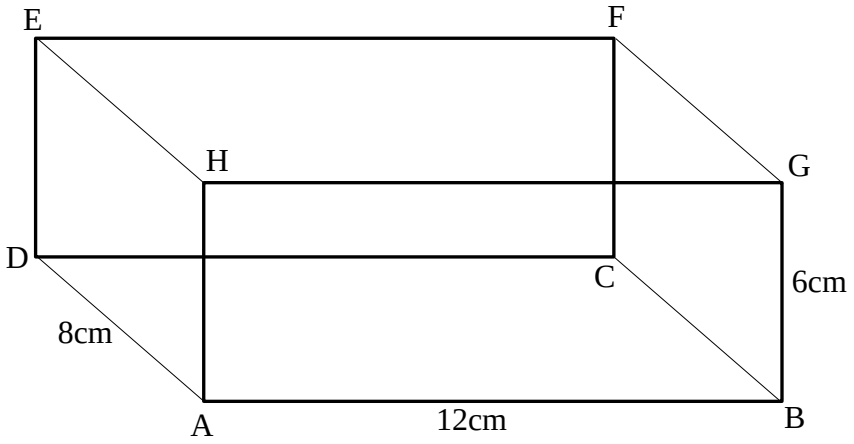
(b) Find the length of a tangent $\overline{RT}$. If $\overline{PQ}$ = 6cm and $\overline{QR}$=10cm.



12. (a) A rectangular prism below measures 12cm long, 8cm wide and 6cm high. Find:

(i) The angle between the edge $\overline{BE}$ and the base

(ii) The total surface area



(b) A bus owned by Semuyenzi Company Ltd leaves town A (3°S, 39°E) at a constant speed of 40km/hr. How many hours will the bus take to reach town B (12°S, 39°E)?

13. (a) Find the image of the point (-2,6) after a reflection in the line y + x = 0

(b) Find the coordinates of the image of the vertices of ΔABC after enlargement by the factor 2 where A(1,1), B(1,2), C(2,1) and hence draw the object and its image on the same xy – plane.

14. (a) Given the function f(x) = $\frac{1}{x-2}$, find:

(i) The domain and range.

(ii) $f^{-1}(\frac{1}{3})$

(b) A farm is to be planted with wheat and maize as shown in the table below:

	Wheat	**Maize**	**Maximum total**
Days of labour per hectare	2	1	10
Labour cost per hectare (Tsh)	700	600	4200
Cost of fertilizer per hectare (Tsh)	300	400	2400

If wheat yields a profit of 800Tsh per hectare and maize yields 600Tsh per hectare, how should the area be planted in order to get a maximum profit?